

MiniMax-H3

From 653.838 Seconds to a Full-VAE Sub-15s Profile

Cumulative optimization · revised from the original 71-page report

14.9692s

Mean complete request

5 / 5

Formal requests below 15s

Full H3 VAE

Original decoder architecture

The new starting point after rollback was 27.4937 seconds. Four-step FastH3, VSA and existing exact engineering stayed in place. The later E4M3-wire / lightweight-decoder branch was retired from the selected path.

The revised story follows the profiling work that came next: gate projection overlap, QKV splitting, Q/K preparation during communication, four-chunk output pipelining, decoder scheduling and allocator changes. These changes are shown separately from new low-precision compute.

Selected result: BF16 communication, MXFP8 DiT and full video VAE, Sage/CAKE attention, and the validated scheduling changes. Timing ends after the complete H.264/AAC MP4 has been saved.

The historical 653.838s result, the old 14.561s endpoint and this revision have different request/configuration contracts. No matched 653.838 / 14.9692 speedup is claimed.

Reading guide

Pages	Contents
2-4	Rollback, retained foundations and the revised cumulative tables
5-9	Measured gains, profile findings and dependency changes
10-12	Quality, final samples, scope and evidence
13-24	Retained technical pages from the original report, labeled historical

What changed after the old report

The old 14.561s result inherited E4M3 QKV transport. Subsequent complete-video human review rejected that transport state. Its timing remains historical evidence, and its recommendation is superseded.

Old choice	Revised selected choice
E4M3 QKV wire	BF16 QKV/O transport. Quantized GEMMs do not imply quantized wire.
TAEH3 lightweight video decoder	Full H3 VAE, patch-parallel on eight ranks; online MXFP8 added later. TAEH3 was retired from this profile, not declared to have failed its own quality gate.
Earlier FP8 linear recipe	MXFP8 E4M3 values and E8M0 block-32 scales in the selected DiT/VAE paths.
Earlier sparse attention kernel	Sage/CAKE approximate attention, with a separately measured later TMA descriptor update.
TAEH3 rank 0 / Audio VAE rank 1	Full video VAE across eight ranks; rank-0 auxiliary-stream Audio VAE joined by an event.

Foundations retained from the original work

Historical checkpoint	E2E / s
Original H3, 49 diffusion forwards	653.838
AdaLN TP1/SP8 + RDMA Ulysses + direct Q/K layout	590.562
Dense FastH3, four forwards	66.785
FastH3 + VSA + FlashInfer	43.420
GPU pixel packing, pinned D2H/encode pipeline, persistent communication and exact VAE/layout cleanup	28.236

These are historical checkpoints under the original prompt. The rollback-era BF16 VSA control was separately measured at 27.493667s. Restoring precision did not discard the existing exact engineering work.

Revised cumulative table · part 1

Start from the restored BF16 VSA / full-VAE control. Each arrow is its own measured comparison; restarted controls are preserved.

Step	Adopted change	E2E / s	Saved / s
Rollback-era baseline	BF16 VSA / linears / wire; full H3 VAE	27.4937	—
Sage/CAKE attention	Sage/CAKE with ragged-tail correction Approximate compute	27.4937 → 25.7063	1.7873
QK/V + four-chunk O	Q/K preparation during V exchange; O return/projection pipeline Byte-exact against that reference	25.7613 → 25.1353	0.6260
Gate projection overlap	Original BF16 gate GEMM on the side stream during Q/K exchange Byte-exact against that reference	25.1177 → 24.3403	0.7773
Views + Q + coarse/fine	Remove layout copies; earlier Q preparation; parallel coarse/fine Byte-exact against that reference	24.3647 → 24.0117	0.3530
QKV projection split	First complete QK + V split A/B; later scheduling refinements inherited Byte-exact against that reference	24.0268 → 23.6654	0.3614
Paired VAE windows	Balance spatial tiles across ranks and overlap assembly Byte-exact against that reference	23.6536 → 22.6670	0.9866
Exact VAE B2 batching	Batch paired tiles; preserve sensitive to_out at B1 Byte-exact against that reference	22.6670 → 22.3662	0.3008
Fixed allocator segments	expandable_segments:False Byte-exact against that reference	22.5924 → 21.7712	0.8212

The 0.3614s split result is the first complete QK + V comparison. Later V-first, gate-after-V and Q/K submission refinements continue that mechanism; their comparisons against fused QKV are not added again.

The earlier 24.0117s endpoint was not reproduced unchanged in every later same-feature cohort. Historical means describe their own runs, not a guaranteed stable staircase.

Revised cumulative table · part 2

Introduce the new DiT precision baseline, then continue through decoder quantization and the final scheduling improvements.

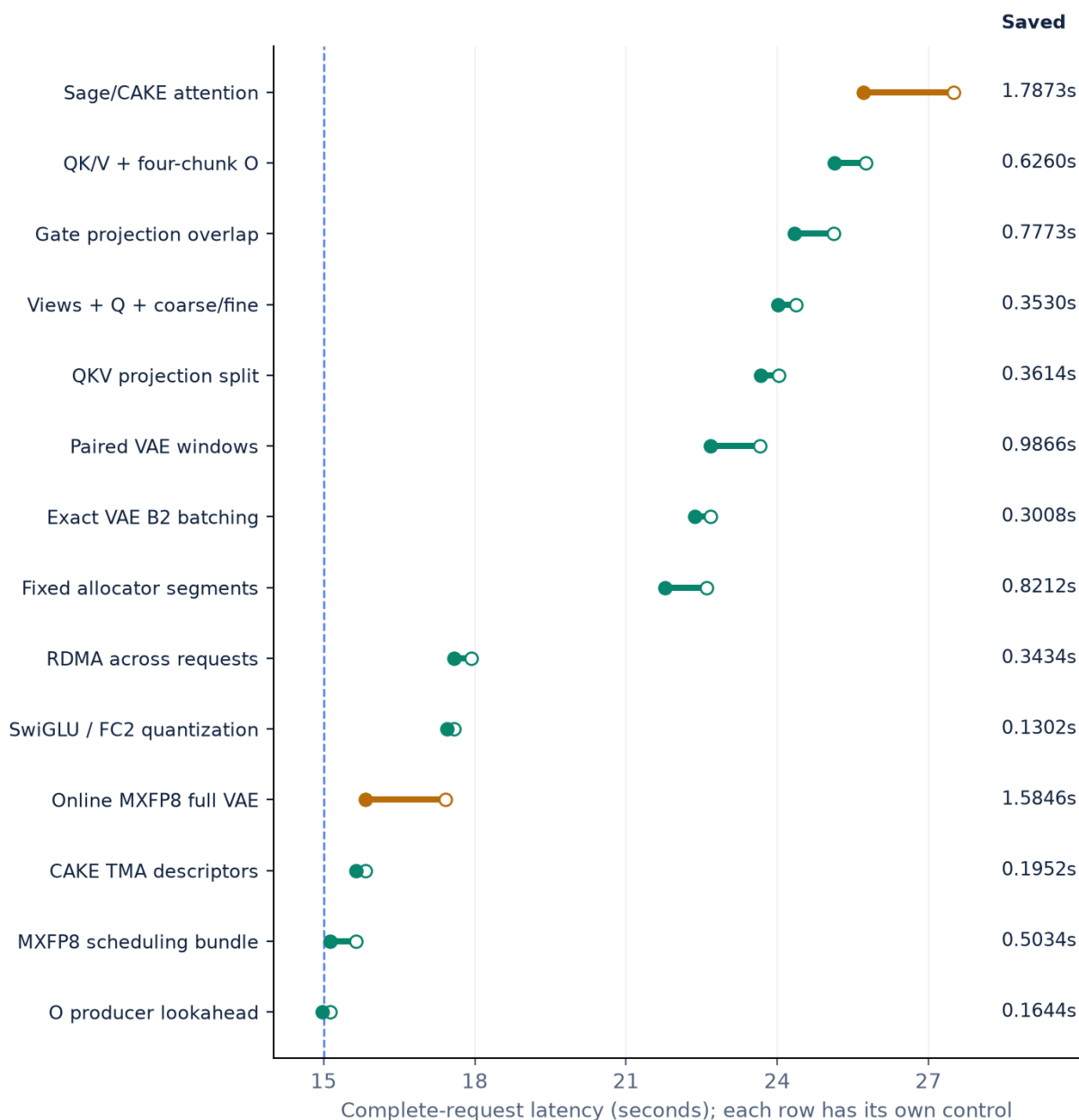
Step	Adopted change	E2E / s	Saved / s
DiT MXFP8 baseline	New low-precision compute and quantization call boundary; BF16 wire and full VAE retained.	17.9240	Unpaired
RDMA across requests	Retain registered communication resources	17.9214 → 17.5780	0.3434
SwiGLU / FC2 quantization	Fuse the activation producer while retaining reference rounding	17.5780 → 17.4478	0.1302
Existing later baseline	Qualified inherited configuration; smaller inherited changes are not separate gain rows.	17.4168	—
Online MXFP8 full VAE	Quantize the full H3 decoder online Approximate compute	17.4168 → 15.8322	1.5846
CAKE TMA descriptors	Descriptor-by-value adaptation on the existing CAKE path Byte-exact against quantized reference	15.8322 → 15.6370	0.1952
MXFP8 scheduling bundle	MXFP8 QK/V split + VAE B4 + audio/video decode overlap Byte-exact against quantized reference	15.6370 → 15.1336	0.5034
O producer lookahead	Prequeue four O packs with per-chunk readiness events Byte-exact against quantized reference	15.1336 → 14.9692	0.1644

No invented precision delta. The 17.9240s DiT MXFP8 experiment changed the quantization call boundary and disabled the old BF16 split. It did not include a matched BF16 control. The difference from the earlier 21.7712s run is not assigned entirely to quantization.

Only adopted additions saving more than 0.100s appear as gain rows. Existing baseline code is not represented as removed or remeasured after removal. The selected endpoint is model-oproducer-r2.

Measured gains across the adopted path

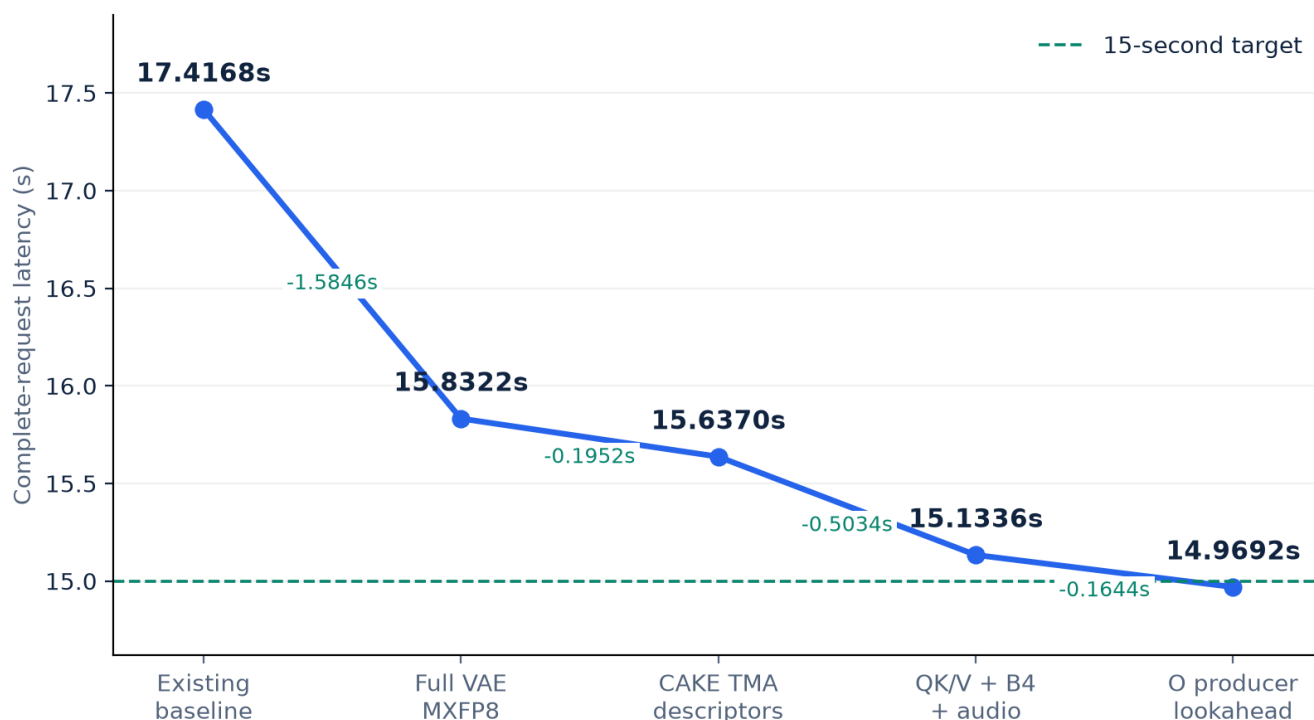
Hollow point: that round's control. Filled point: candidate. Green preserves the selected reference bytes; amber changes numerical precision.



This comparison chart preserves the actual controls instead of fabricating one uninterrupted sweep. The separate DiT MXFP8 checkpoint has no matched BF16 delta and is intentionally absent from the gain plot.

The final cumulative descent

From the existing MXFP8 DiT / full H3 VAE baseline to the selected complete-request endpoint.



2.4476s

Saved from 17.4168s

0.8630s

Byte-exact tail after VAE MXFP8

Online full-VAE MXFP8: 1.5846s. Original FP32 weights are quantized online to E4M3 values with E8M0 block-32 scales. Dynamic activations, FP16 output and 144 converted layers per worker retain the full decoder architecture.

CAKE descriptors: 0.1952s. The measured change is the TMA descriptor ABI on an already specialized CAKE path. The H3 specialization itself has no isolated gain in this table.

Scheduling bundle: 0.5034s. MXFP8-aware QK/V splitting, B4 decoder scheduling and audio/video overlap were validated together. Their individual contributions were not isolated.

O producer lookahead: 0.1644s. Prequeuing the four pack operations removes host submission dependencies while preserving per-chunk readiness and the original transport protocol.

Five-sample sequential cohorts support these observations. The plot is not a randomized interleaved study, and local kernel savings are not added to the complete-request measurements.

What the profiles revealed

The pipeline changes were developed from actual Nsight timelines and data dependencies. A faster attention kernel alone would leave these serialized operations in place.

Observed serialization	Implemented schedule	Measured evidence
Q/K preparation waited for V exchange.	Prepare Q/K on the side stream while communication continues with V.	QK/V + four-chunk O bundle: 0.6260s E2E.
The VSA gate projection ran before QKV exchange.	Run the original BF16 gate GEMM on the side stream during Q/K communication.	Gate-only A/B: 0.7773s E2E.
All O data returned before projection.	Return four 2998-row chunks; gate/project each ready chunk during later transfers.	GPU7 reverse-path example: 8.626 → 6.754ms.
Q/K/V layout materialization, late Q work and serialized coarse attention.	Use strided views, prepare Q earlier and run coarse/fine concurrently with an explicit join.	Three-change bundle: 0.3530s E2E.

Gate projection: a concrete profile example

On GPU7, step 3 / layer 0, the original gate GEMM took about **4.536ms**; **4.473ms** overlapped the Q exchange interior after rescheduling. The QKV-start-to-V-closing-start span shortened **29.115 → 25.997ms**.

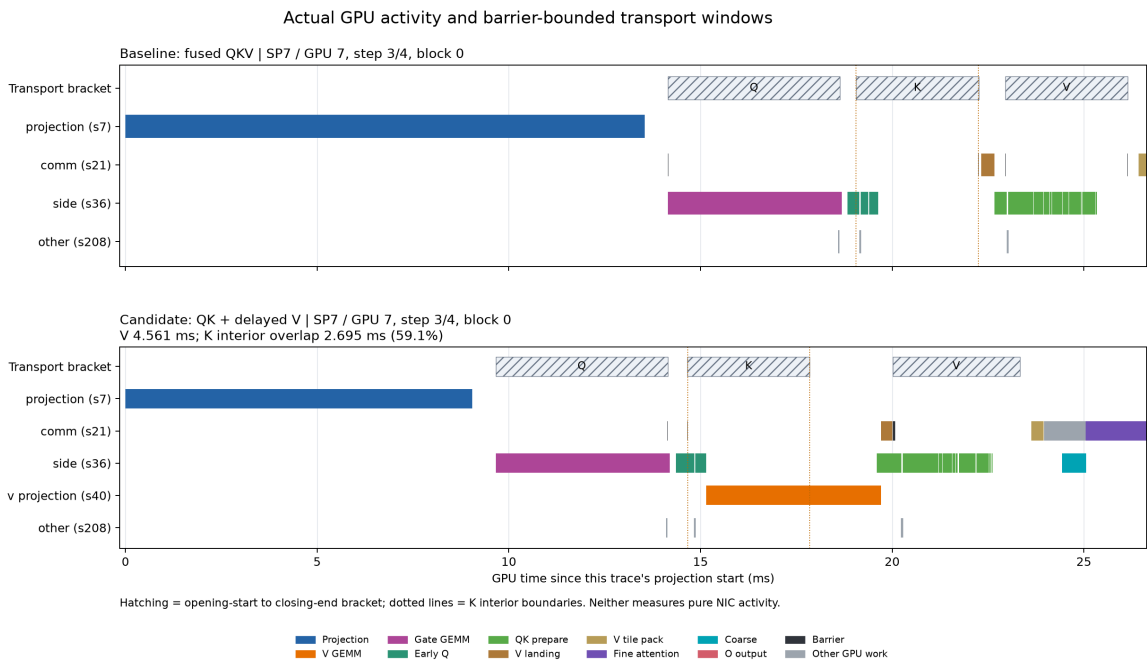
That gate depends on the original hidden input, not on remote Q/K/V. Its shape and BF16 rounding remain unchanged. The output-side elementwise gate and the MLP SwiGLU are different operations.

Exchange brackets include synchronization and scheduling waits. These overlaps are not measurements of NIC payload-active time. The final O projection tail remains exposed; the report does not claim all communication or computation is hidden.

Source: [original pipeline profile breakdown and per-layer evidence](#).

QKV splitting opens another overlap window

Preparation overlap and projection splitting are different changes. The first moves already available Q/K processing under V communication. The second splits the producer GEMM into QK and V so V computation can overlap Q/K transport.



Actual historical Nsight figure: GPU7, step 3 / layer 0, fused QKV versus the first QK + delayed-V split. This is a captured timeline, not a schematic or the final MXFP8 trace.

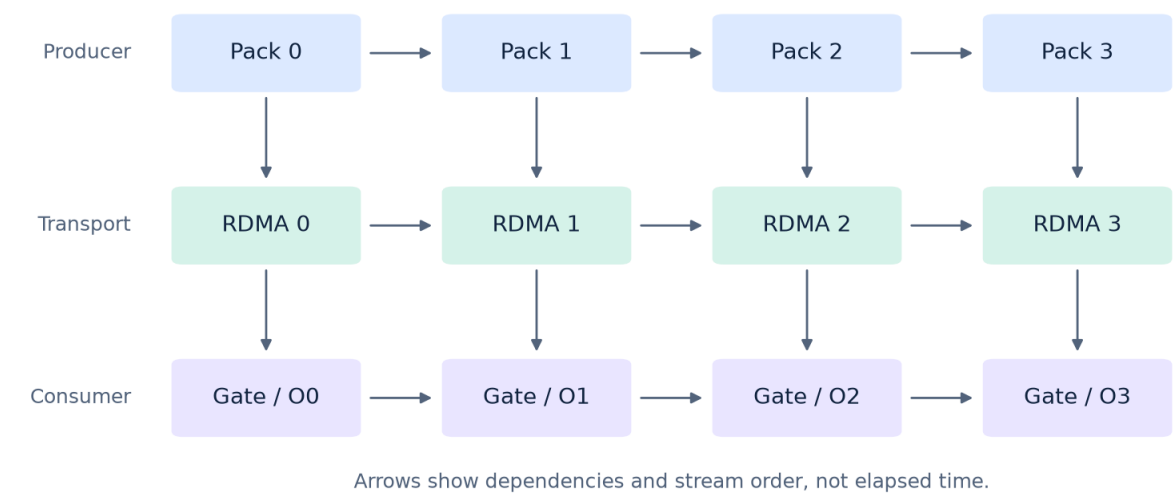
First split qualification	Result
Complete requests	24.0268 → 23.6654s; 0.3614s observed mean saving
Numerical/trace evidence	Ten formal MP4s match; 400 rank/block dependency checks in each trace
Remaining exposure	GPU7 V averaged 4.5626ms; 2.7565ms overlapped K interior, with 1.7297ms of V activity after K close

The subsequent schedule launched V before Q exchange, made the gate wait for V, and moved Q/K preparation into the V transfer window. This refines the same mechanism and is not an extra additive split gain.

The old BF16 split bypassed the newer quantization interface. The final path therefore reimplemented QK/V splitting for MXFP8 and checked actual projection bytes and full videos before admission.

Remove false dependencies; preserve real ones

The final O improvement keeps four original chunks and four original exchanges. It changes when pack operations enter the producer stream.



Before: return chunk 0, then submit pack 1; return chunk 1, then submit pack 2. Those host-order edges delayed work that did not depend on the previous return.

After: queue all four packs before the first blocking exchange. Each RDMA still waits for its own pack-ready event. Separate slots, original exchange order and the consumer join preserve data visibility and buffer lifetime.

Complete-request result: **15.1336 → 14.9692s**. The gain is 164.4ms; it is not converted into an inferred per-stage overlap percentage.

The decoder and output pipelines

Adopted mechanism	Purpose
Paired temporal windows + exact B2	Balance the eight VAE ranks and reduce repeated decode overhead while preserving reference bytes.
MXFP8 B4 + Audio VAE overlap	Use the qualified B4/B1 plan; launch rank-0 audio on an auxiliary stream before video decode and join before its consumer.
Pinned D2H / CPU encoding	Overlap output copies and encoding; the request still waits for the complete MP4.
Resident RDMA + fused SwiGLU/quantization	Avoid repeated registration and an intermediate activation producer without changing its reference values.

Quality: separate approximation from identity

0.97892

SSIM, all 362 frames

43.0414 dB

PSNR, all 362 frames



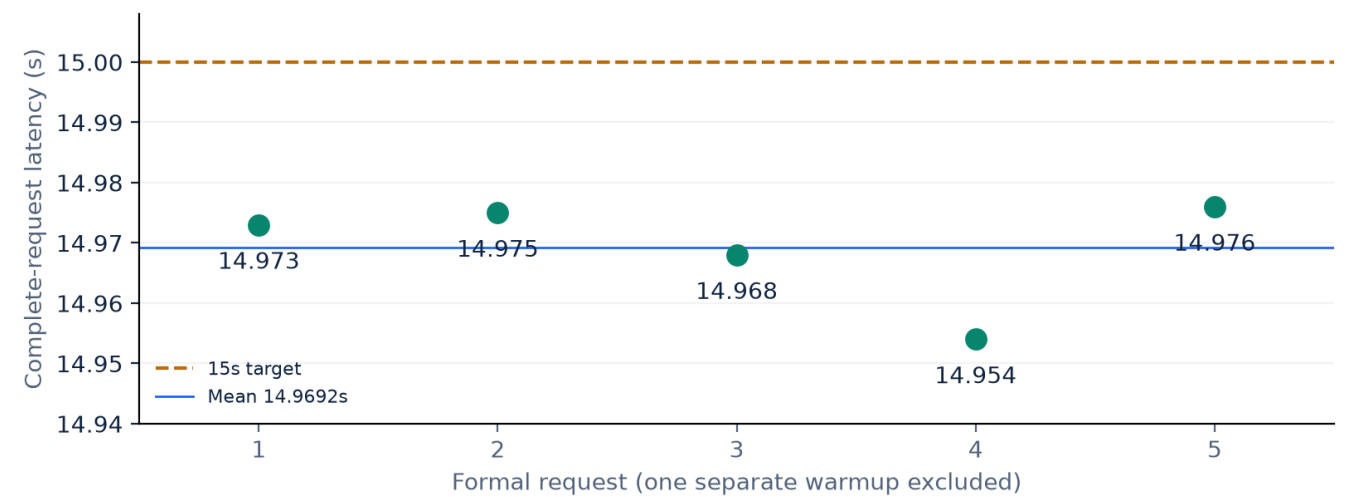
Same decoded frames at 1.0, 4.5, 9.0 and 14.0s. Top: full H3 VAE before MXFP8. Bottom: full H3 VAE with MXFP8; this candidate MP4 is byte-identical to the final selected formal video.

Check	Scope and result
VAE precision boundary	All 362 decoded MP4 frames, including codec effects; audio PCM identical. The metrics describe one formal-seed video, not all prompts.
Scheduling output identity	Six complete MP4s match their corresponding references: one warmup plus five formal requests. These are not six independent prompts/seeds.
Actual execution checks	400 QKV byte gates, 400 O byte gates, 48 gather markers and 12 audio start/join events.
Whole-path precision	FastH3, VSA, Sage/CAKE and MXFP8 are approximate. Byte identity applies to the validated engineering changes after selecting the numerical reference.

Full VAE and BF16 wire remove the old decoder/wire choices. Neither PSNR/SSIM nor the screenshots establish universal perceptual equivalence to the original dense BF16 model.

[Reference MP4](#) · [Selected-output MP4](#)

Final requests and measurement contract



14.9692s
Mean; sample SD 0.00904s

14.976s
Slowest of five requests

30.8ms
Mean margin to 15s

Contract	Observed configuration
Timing	Client HTTP POST start to complete MP4 saved; model load, warmup and post-run validation excluded. Concurrency 1; profiler disabled.
Media	362 frames, 24 fps, effective 1280×704 from a 1280×720 request; H.264 + stereo 32kHz AAC; formal seed 1101.
Model / parallelism	FastH3 VSA/Data-Free, four forwards, top-k 162, tile 64. DiT TP1/Ulysses8/Ring1; text encoder TP8; full video VAE parallel8.
GPU / network	8× SM120; recorded device name NVIDIA Graphics Device, 73415MiB each. Four ConnectX-8 groups / eight active 400Gb/s mlx5 RDMA devices.
Runtime	Driver 580.95.05; PyTorch 2.13.0+cu132; vLLM 0.28.0; OMP 28; model resident; node-default unlocked clocks.
Run integrity	4,784 source files pinned; source unchanged; zero formal foreign-process samples. Five sequential requests, not an independent interleaved replication.

The measured five requests meet the strict target. The 30.8ms margin is specific to this fixed workload and node; it is not a latency guarantee for other prompts, loads or two-NIC machines.

Evidence, provenance and retained pages

The revised PDF replaces the original download at the same path. The original 71-page edition is archived unchanged. Its E4M3-dependent 14.561s recommendation is superseded by the later quality review.

Artifact	What it supports
adopted-cumulative.zh.md	Chinese cumulative table, stage descriptions and source paths
data.json + evidence/source-manifest.json	Selected request/quality values, original source hashes and six historical evidence snapshots
evidence/pipeline-profile-breakdown.md	Actual gate, QK/V, O and coarse/fine timeline findings
evidence/qkv-split-results.md	Initial QK + V full-model timings and trace qualification
figures/*.svg and *.png	Exportable charts generated from the recorded values
../quality/moon-teahouse-seed1101/README.md	Subsequent complete-video rejection of E4M3 QKV transport
../archive-20260907/	Unchanged original PDF, data and generator; historical archive

Selected formal MP4 SHA-256

9e6a9cd878b689ed618331ca935cd83f35e8c74aad5d8311d17438717c943307

Selected source-map SHA-256

7411847da54842dedd32105de66ab79573ffd291e85f57a37946e6de4a7478a8

Historical prompt: 613e0a1beccf83566e3531852c52c2e11daffa3b255222cd67726cdf73c1a563

Current prompt: 86a513458a785fb9b3cada9289c0336e99ff7a26ad8eaf5dda593784cd7bf288

Retained original technical pages

Pages 13–24 retain original pages 8, 15–24 and 26: original architecture, AdaLN placement, RDMA Ulysses, direct layouts, FastH3, VSA, output processing, persistent communication, packing and exact VAE cleanup. Their values and any quality-section references belong to the historical edition.

The old rejected endpoint, unrelated unsuccessful branches and sub-100ms additions are excluded from this edition's adopted-gain narrative. No GPU experiments were run to create this revision.

[Report landing page](#) · [Current data](#) · [Original archive](#)

Original architecture: where 653.838 seconds went

Original 653.838 s path



The original-H3 control used 49 diffusion forwards and a collective-heavy full video VAE.

DiT alone consumed 633.058 s. The first accepted optimizations attacked distributed overhead without changing the original model. Nsight later showed cuDNN attention as 64.53% of summed GPU kernel time in the profiled step, followed by GEMM at 19.82% and RDMA-barrier residency at 14.36%.

Optimization 01 — S1: AdaLN TP1 / SP8

653.838 s -> 605.348 s | measured reduction 48.490 s | cumulative speedup 1.080x

Dimension	Evidence-backed statement
Phase	A
Status	accepted
Fidelity	Exact
Principle	Replicate small AdaLN tensors and spend all eight ranks on sequence parallelism.
Evidence	DiT 584.707 s; -48.490 s E2E.
Interpretation	Removes tensor-parallel collectives from a latency-dominated batch-one request.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 02 — S2: Explicit RDMA Ulysses exchange

605.348 s -> 592.079 s | measured reduction 13.269 s | cumulative speedup 1.104x

Dimension	Evidence-backed statement
Phase	A
Status	accepted
Fidelity	Exact
Principle	Move the large sequence exchange through the topology-aware RDMA path.
Evidence	DiT 571.264 s; -13.269 s E2E.
Interpretation	The gain is interconnect-path selection, not model approximation.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 03 — S3: Direct Q/K layout

592.079 s -> 590.562 s | measured reduction 1.517 s | cumulative speedup 1.107x

Dimension	Evidence-backed statement
Phase	A
Status	accepted
Fidelity	Exact
Principle	Produce the attention layout directly and delete redundant transposes/copies.
Evidence	DiT 569.830 s; -1.517 s E2E.
Interpretation	A small but repeatable layout win before distillation.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 04 — F0: Dense FastH3, four forwards

590.562 s -> 66.785 s | measured reduction 523.777 s | cumulative speedup 9.790x

Dimension	Evidence-backed statement
Phase	B
Status	accepted
Fidelity	Lossy
Principle	Use the distilled four-step sigma schedule [0.999, 0.749, 0.5, 0.25, 0].
Evidence	DiT 46.988 s; -523.777 s E2E.
Interpretation	This is the largest single change and the first explicit quality boundary.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This optimization crosses a declared approximation boundary. The dedicated quality section contains screenshots and full-video post-codec metrics for the nearest available matched artifact pair.

Optimization 05 — P0: FastH3 + VSA + FlashInfer

66.785 s -> 43.420 s | measured reduction 23.365 s | cumulative speedup 15.058x

Dimension	Evidence-backed statement
Phase	B
Status	accepted
Fidelity	Lossy
Principle	Apply visual sparse attention with exact FlashInfer sparse-kernel execution inside the distilled trajectory.
Evidence	DiT 22.704 s; -23.365 s E2E.
Interpretation	VSA changes the attention graph; the kernel itself is exact for that sparse graph.

Causality boundary

This rung contains a measured bundle. The E2E delta is attributed to the bundle only; component-level claims come from separate microbenchmarks or profiler evidence and are not summed into a fictional decomposition.

Quality handling

This optimization crosses a declared approximation boundary. The dedicated quality section contains screenshots and full-video post-codec metrics for the nearest available matched artifact pair.

Optimization 06 — P1: GPU FP32-to-uint8 video pack

43.420 s -> 36.568 s | measured reduction 6.852 s | cumulative speedup 17.880x

Dimension	Evidence-backed statement
Phase	C
Status	accepted
Fidelity	Exact
Principle	Quantize display pixels on GPU and copy uint8 instead of float frames.
Evidence	-6.852 s E2E.
Interpretation	Semantics are unchanged at the MP4 contract boundary.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 07 — P2: Chunked pinned D2H + background mux

36.568 s -> 31.399 s | measured reduction 5.169 s | cumulative speedup 20.824x

Dimension	Evidence-backed statement
Phase	C
Status	accepted
Fidelity	Exact
Principle	Overlap pinned host copies and CPU PyAV encode/mux with remaining device work.
Evidence	-5.169 s E2E.
Interpretation	The POST endpoint still waits for a valid H.264/AAC MP4.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 08 — P3: Persistent RDMA communication

31.399 s -> 30.534 s | measured reduction 0.865 s | cumulative speedup 21.413x

Dimension	Evidence-backed statement
Phase	C
Status	accepted
Fidelity	Exact
Principle	Reuse communication resources across layer iterations instead of reinitializing them.
Evidence	-0.865 s E2E.
Interpretation	Reduces orchestration and allocator overhead.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 09 — P4a: Fused VSA tile packing

30.534 s -> 30.013 s | measured reduction 0.521 s | cumulative speedup 21.785x

Dimension	Evidence-backed statement
Phase	C
Status	accepted
Fidelity	Exact
Principle	Fuse tile selection and packing while preserving selected values and order.
Evidence	Approximately -0.490 s in matched trials.
Interpretation	A kernel-launch and memory-traffic optimization.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 10 — P4c: Compact VSA until

30.013 s -> 29.880 s | measured reduction 0.133 s | cumulative speedup 21.882x

Dimension	Evidence-backed statement
Phase	C
Status	accepted
Fidelity	Exact
Principle	Write only the compact valid output domain during sparse-attention reconstruction.
Evidence	-0.133 s E2E.
Interpretation	Avoids materializing padding that no downstream operator consumes.

Causality boundary

The delta is the direct accepted-parent comparison. It is not combined with side-branch deltas.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.

Optimization 12 — P5: Exact VAE/direct Q-to-K cleanup

29.873 s -> 28.236 s | measured reduction 1.637 s | cumulative speedup 23.156x

Dimension	Evidence-backed statement
Phase	C
Status	accepted
Fidelity	Exact
Principle	Consolidate exact decoder and Q-to-K dataflow cleanups.
Evidence	Q0 control: DiT 21.347 s.
Interpretation	A mixed cleanup bundle; it is not used to assign per-kernel causality.

Causality boundary

This rung contains a measured bundle. The E2E delta is attributed to the bundle only; component-level claims come from separate microbenchmarks or profiler evidence and are not summed into a fictional decomposition.

Quality handling

This is classified as an exact systems/dataflow optimization inside the already-selected model and precision policy. Exactness is checked with tensor/media hashes where the corresponding artifact exists.